MTH 5000, Lecture 3

Applied Forecasting of Financial Data
Topic: Accessing and Retrieving Financial Data

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Why Financial Data Matters

Financial data is the foundation of modern economic and business decisions. It helps organizations and individuals understand markets, manage risk, and make informed choices.

It matters because it enables:

- Better decision-making
- Risk management
- Market transparency
- Economic forecasting

In short Financial data turns market activity into insight, and insight into action.

Types of Financial Data

Market Data:

Price and trading information such as open, close, volume, bid/ask, and volatility.

Fundamental Data:

Company financial statements, earnings, revenue, assets, liabilities, and ratios.

Economic Data:

Macroeconomic indicators like GDP, inflation, unemployment, and interest rates.

Event Data:

Discrete events that impact markets, such as earnings announcements, mergers, policy changes, or crises.

Use Cases of Financial Data

- **Stock price prediction:** Uses historical prices, volume, and indicators.
- **Portfolio optimization:** Uses returns, volatility, and correlations between assets.
- **Fraud detection:** Uses transaction patterns, outliers, and behavioral data.
- **Market sentiment analysis:** Uses news articles, social media, and headlines.
- **Risk modeling:** Uses economic data, asset prices, and event data.

What Does Accessing Financial Data Mean?

Accessing financial data means obtaining structured information about markets, companies, and the economy from external sources so it can be analyzed and used for decision-making.

This process turns raw financial information into usable datasets.

Methods to Access Data

Free Sources:

- Yahoo Finance
- Alpha Vantage
- FRED (Federal Reserve)
- SEC EDGAR

Paid Sources:

- Bloomberg
- Refinitiv
- Morningstar
- S&P Global

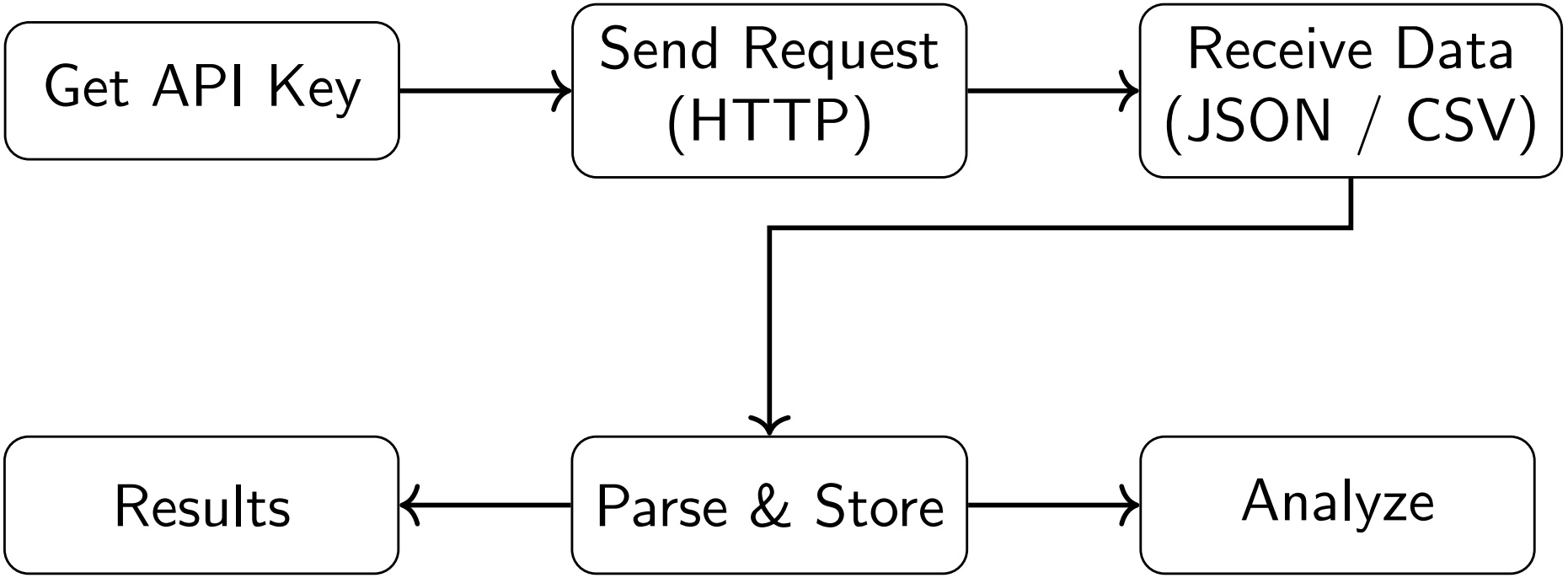
Web APIs

For this project, we will source our data from popular Web APIs, which provide programmatic access to real-time and historical financial information.

- Allow automated data retrieval over the internet
- Return data in structured formats (JSON, CSV, XML)
- Support real-time and historical market data
- Require authentication using API keys
- Are scalable and easy to integrate with Python and other tools

Web API's

How to Use APIs



Challenges and Limitations of APIs

Rate Limits

Most free APIs restrict how many requests you can make per minute or per day, which can slow down large data collection tasks.

Data Quality

APIs may contain missing values, delays, or inconsistent formatting, requiring extra cleaning and validation.

Access & Cost

High-quality financial data is often behind paywalls, and premium tiers can be expensive.

Reliability

APIs can go offline, change formats, or update endpoints, which may break existing code.

Sector Selection Vote

Each student will select ONE stock from the chosen sector. Please vote for ONE option.

1. Technology
Examples: AAPL, MSFT, AMZN, GOOGL, META
Best for: Strong interdependence, AI, cloud computing, innovation-driven relationships

2. Semiconductors
Examples: NVDA, AMD, INTC, QCOM, AVGO
Best for: Extremely high correlation due to shared hardware and supply chain dependencies

3. Finance / Banking
Examples: JPM, BAC, GS, MS, WFC
Best for: Macroeconomic relationships, interest rate sensitivity, financial system analysis

4. Energy
Examples: XOM, CVX, COP, SLB, MPC
Best for: Commodity-driven relationships and dependence on oil price movements

Project Goal

Objective: Build a unified financial dataset for interdependent analysis and forecasting

Each student will:

- Select one stock from the chosen sector
- Acquire historical financial data using Python
- Clean and validate their dataset
- Prepare the data for integration

As a group, you will:

- Combine individual datasets into one unified dataset
- Analyze relationships between stocks
- Identify interdependence and correlations
- Develop forecasting models using the combined data

Outcome: A complete, research-quality dataset ready for interdependent modeling and forecasting